

- 1 (a) Define acceleration.

.....
 [1]

- (b) A rocket is launched vertically from the surface of the Earth.

Fig. 1.1 shows the variation of the velocity of the rocket with time for the first 20 s after its launch.

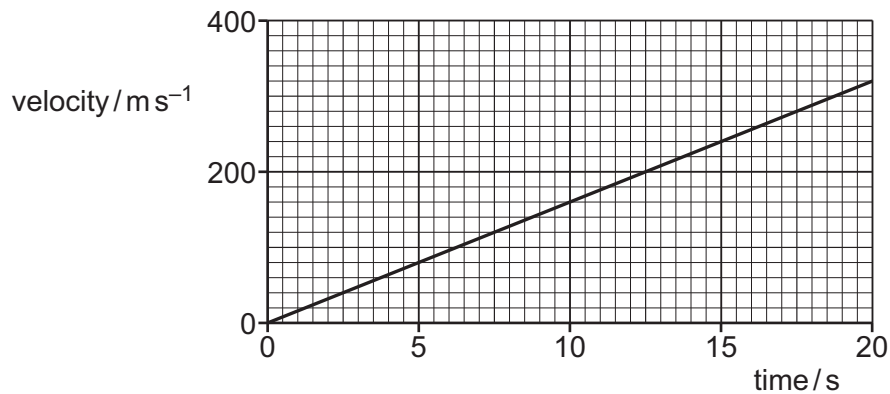


Fig. 1.1

- (i) Determine the acceleration of the rocket.

acceleration = m s^{-2} [1]

- (ii) Show that the height of the rocket above the surface of the Earth at a time of 20 s after launch is 3.2 km.

[2]

